ML2 2.1

Unsupervised Learning Algorithm

Dendrograms and PCA

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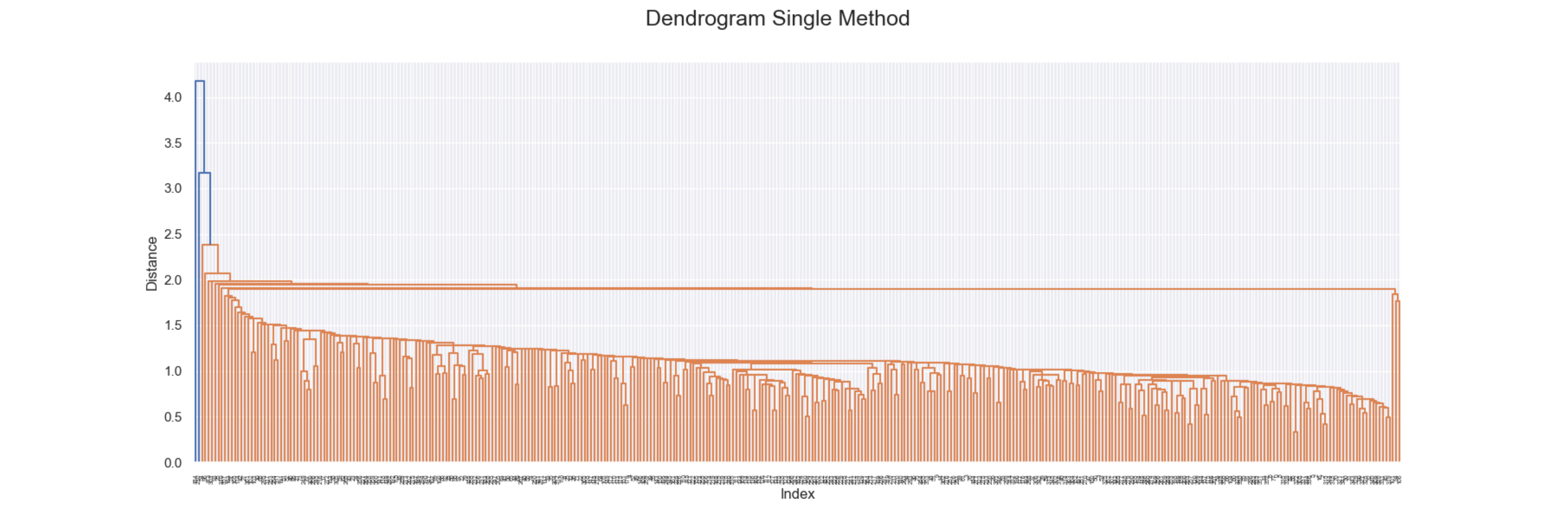
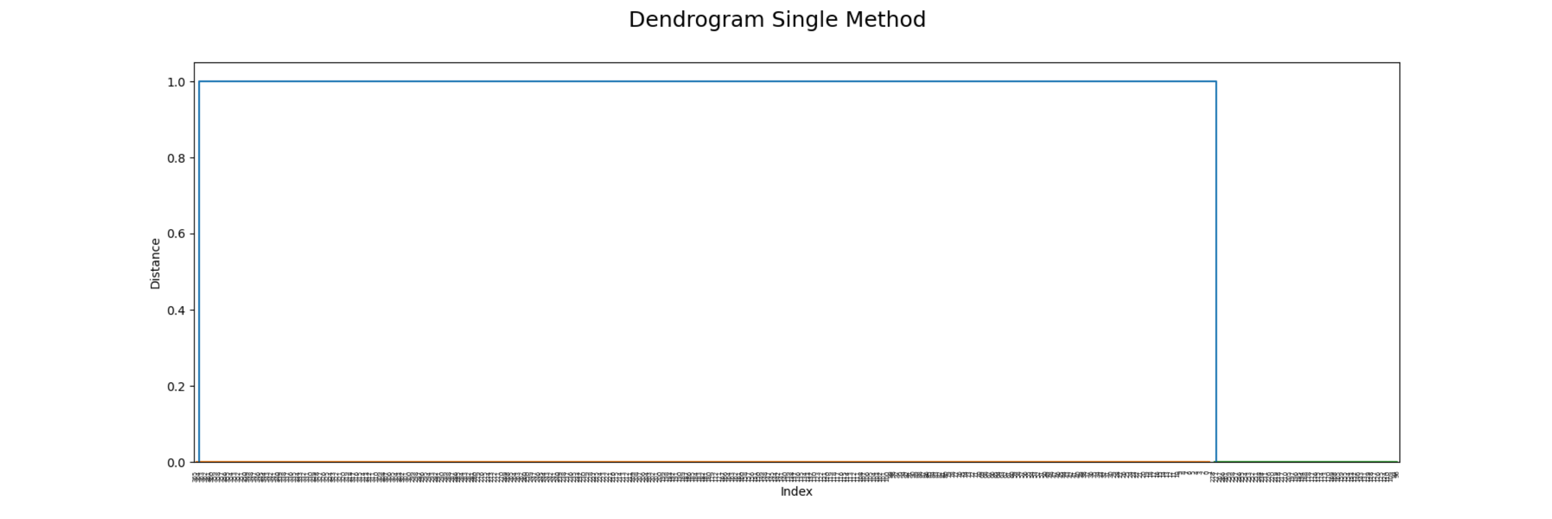
Dendrogram: Single

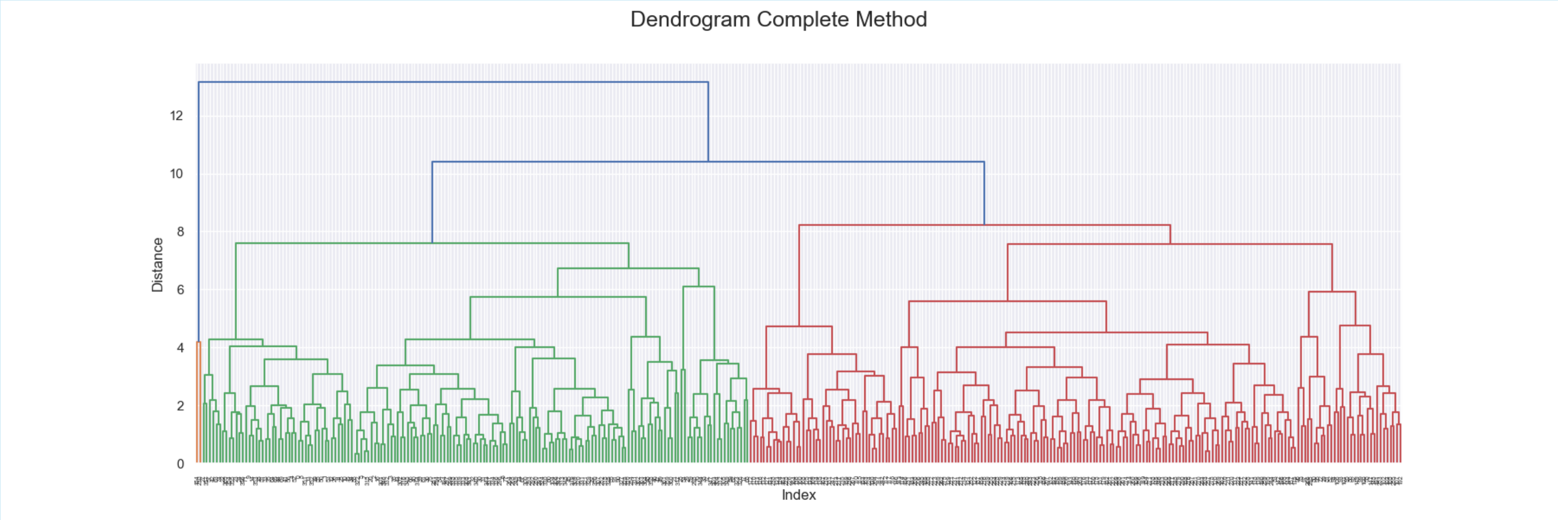
When examining the dendrograms we can see that the single appears to have only 2 variants, and the average also has 2 variants. The ward dendrogram provides better separation.

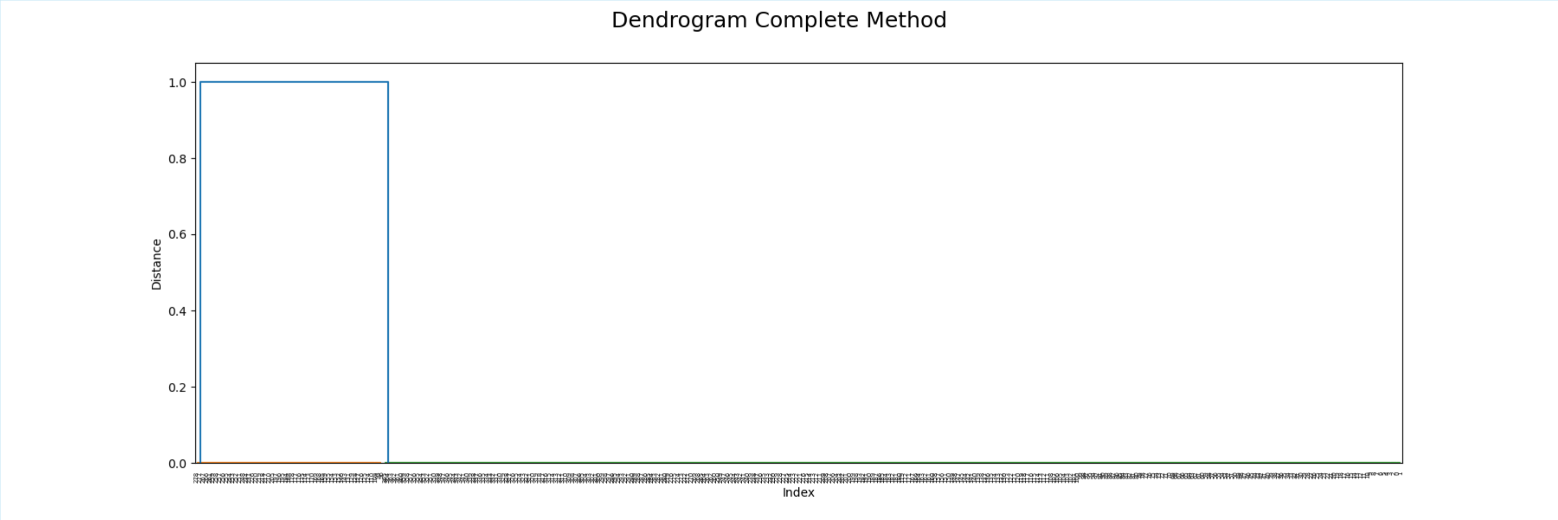




Dendrogram: Complete

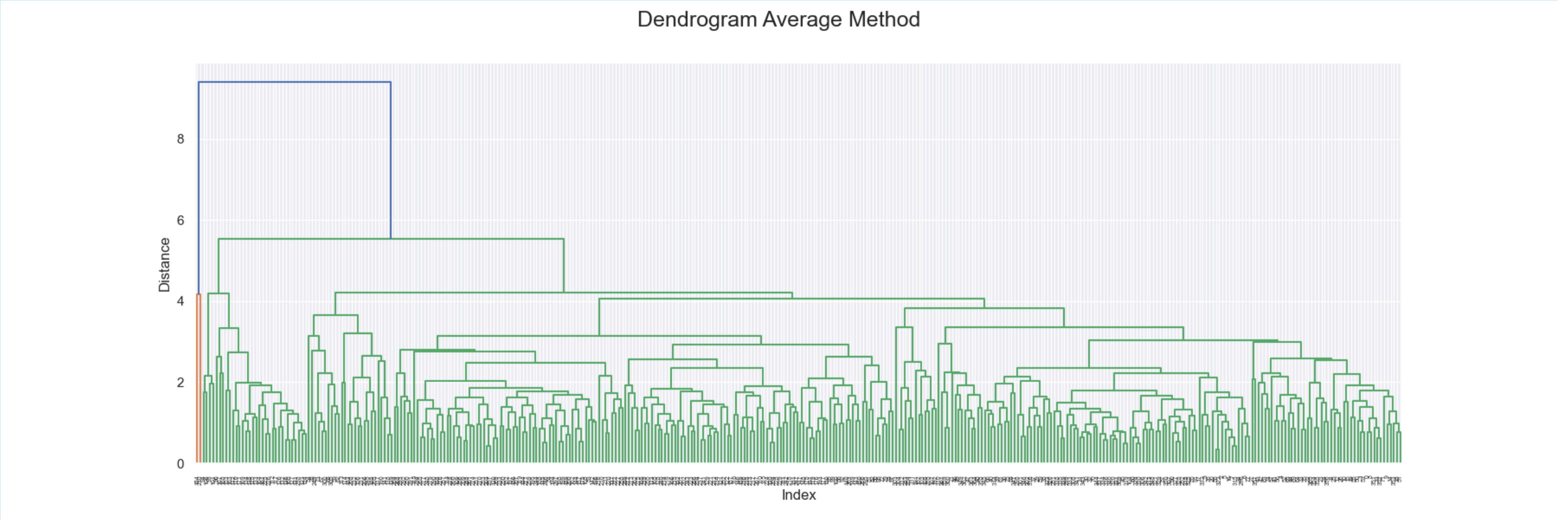
The complete and ward dendrogram have more variation in their decisions.

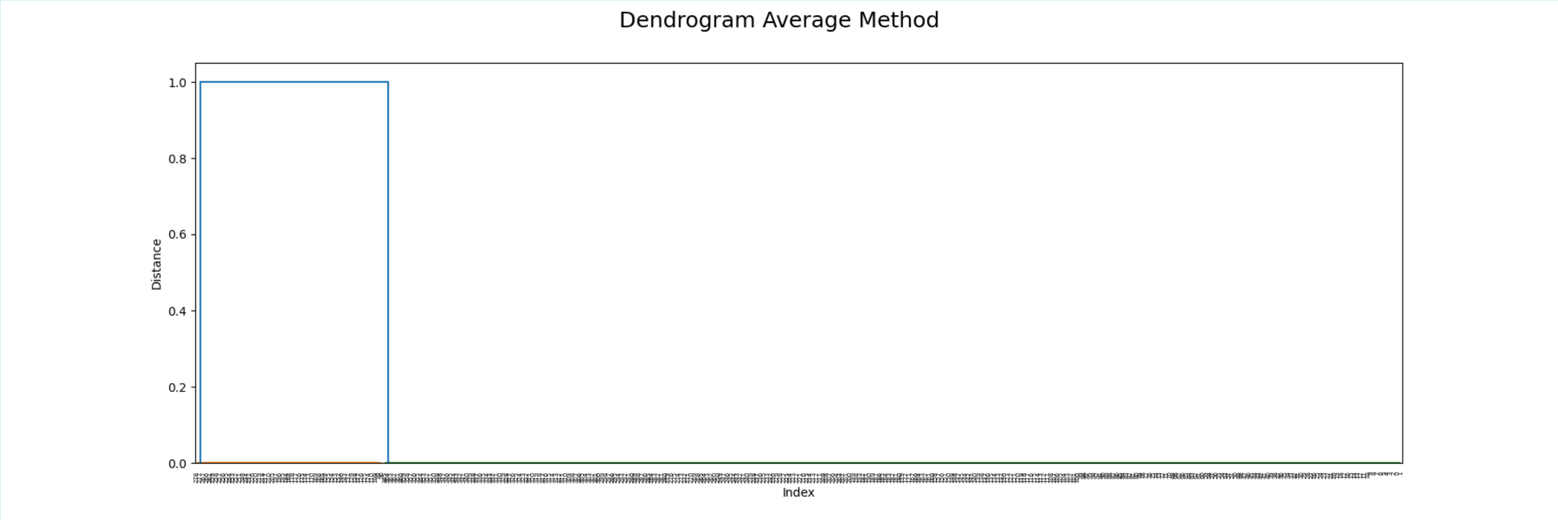




Dendrogram: Average

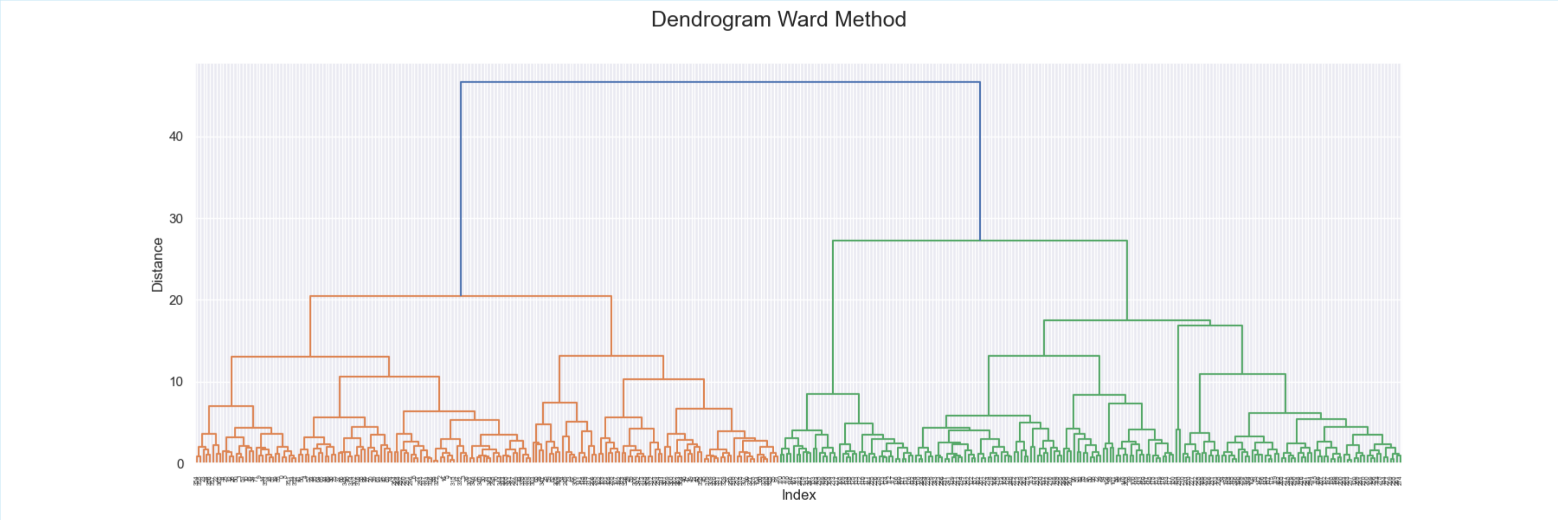
The average dendrogram appears to have 3 sections but it's harder to determine the definitive sections.

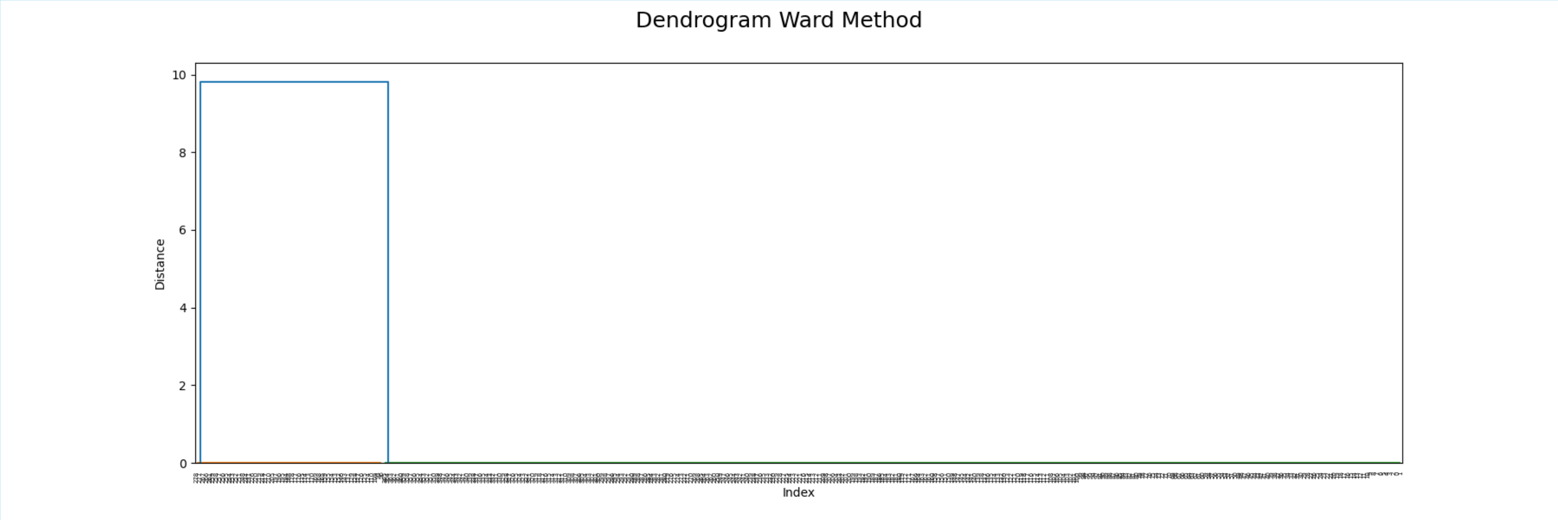




Dendrogram: Ward

The ward dendrogram provides better separation.





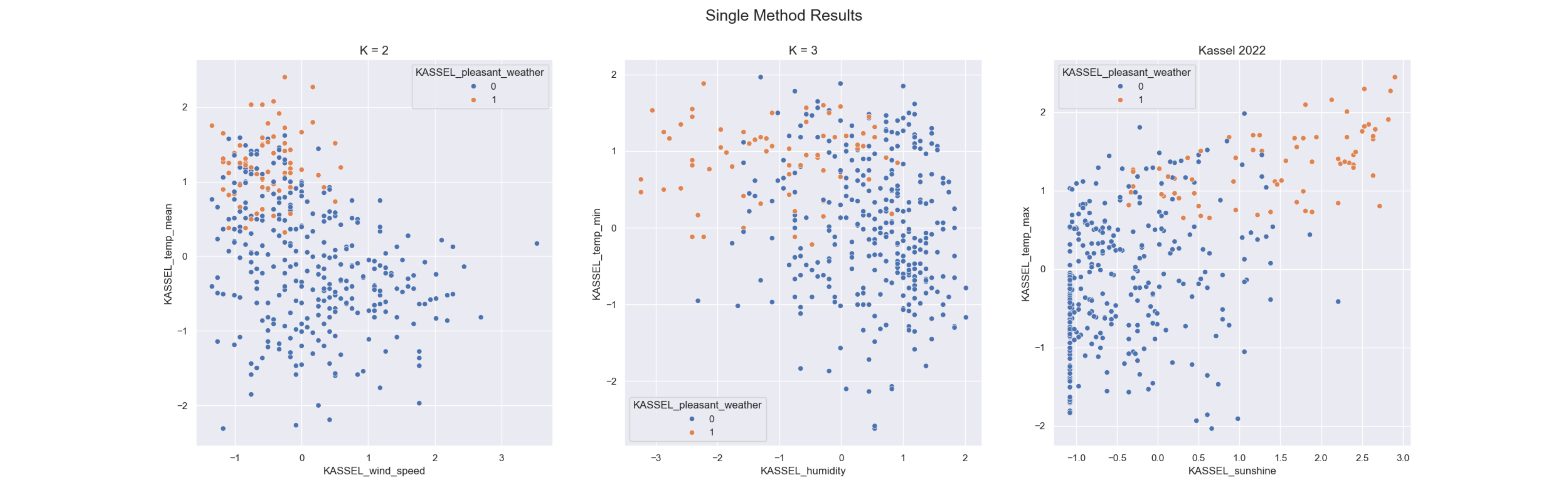
Scatterplot

The pleasant weather is represented by 0 = unpleasant and 1= pleasant. The first scatterplot comparing the Mean Temps and Wind Speed shows that when the wind speeds are low and the mean temps are between the scaled values of 0-3 most dats are pleasant.

The second scatterplot comparing Minimum Temps and Humidity shows that the less humidity and the higher the minimum temperature the more pleasant the day.

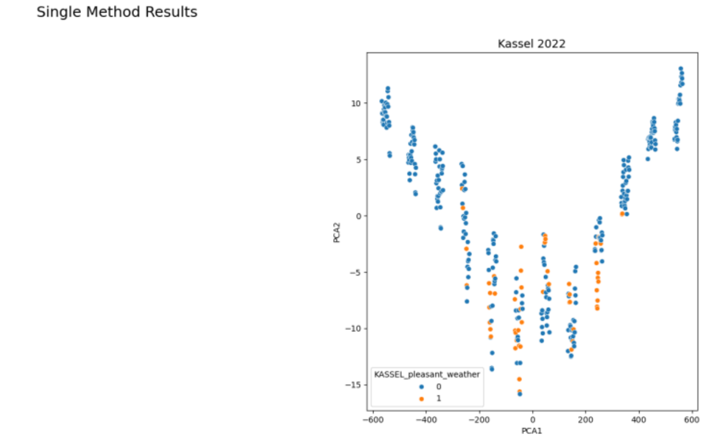
The third scatterplot comparing Max temps and Sunshine show that the sunnier the day with a max temp that falls within the scaled values of 0.5-3 are more pleasant

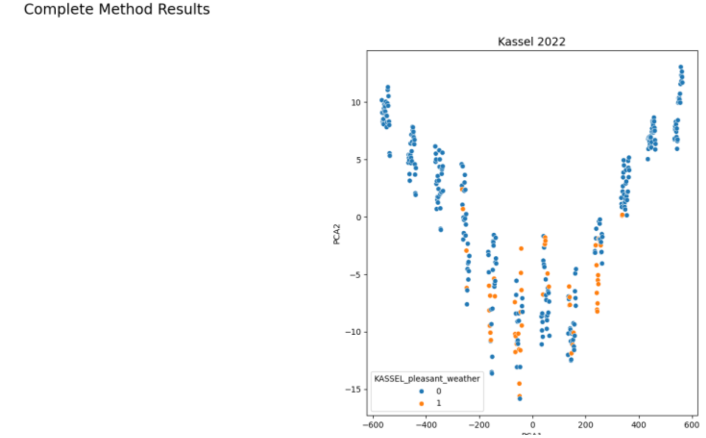
The scatterplots were all the same in the original data when going through single, complete, average, and ward.

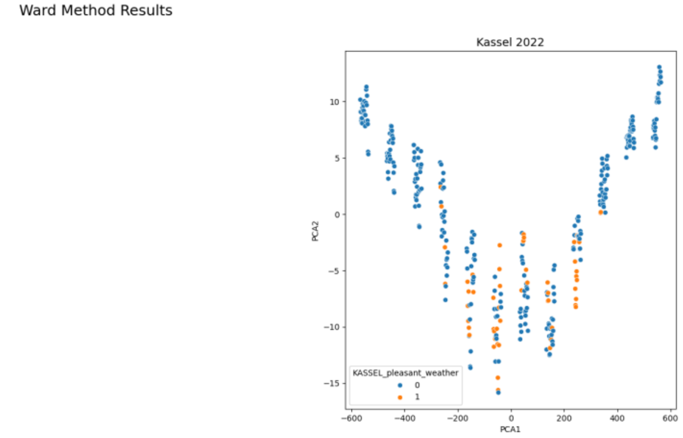


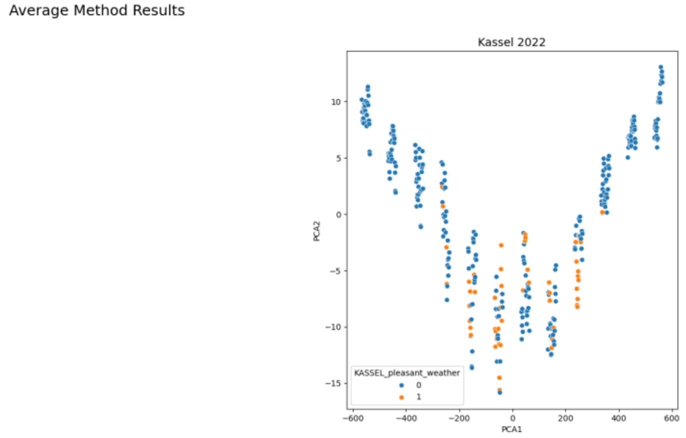
Scatterplot:PCA

When using the PCA reduced data these scatterplots were all identical. They did not provide any insight.









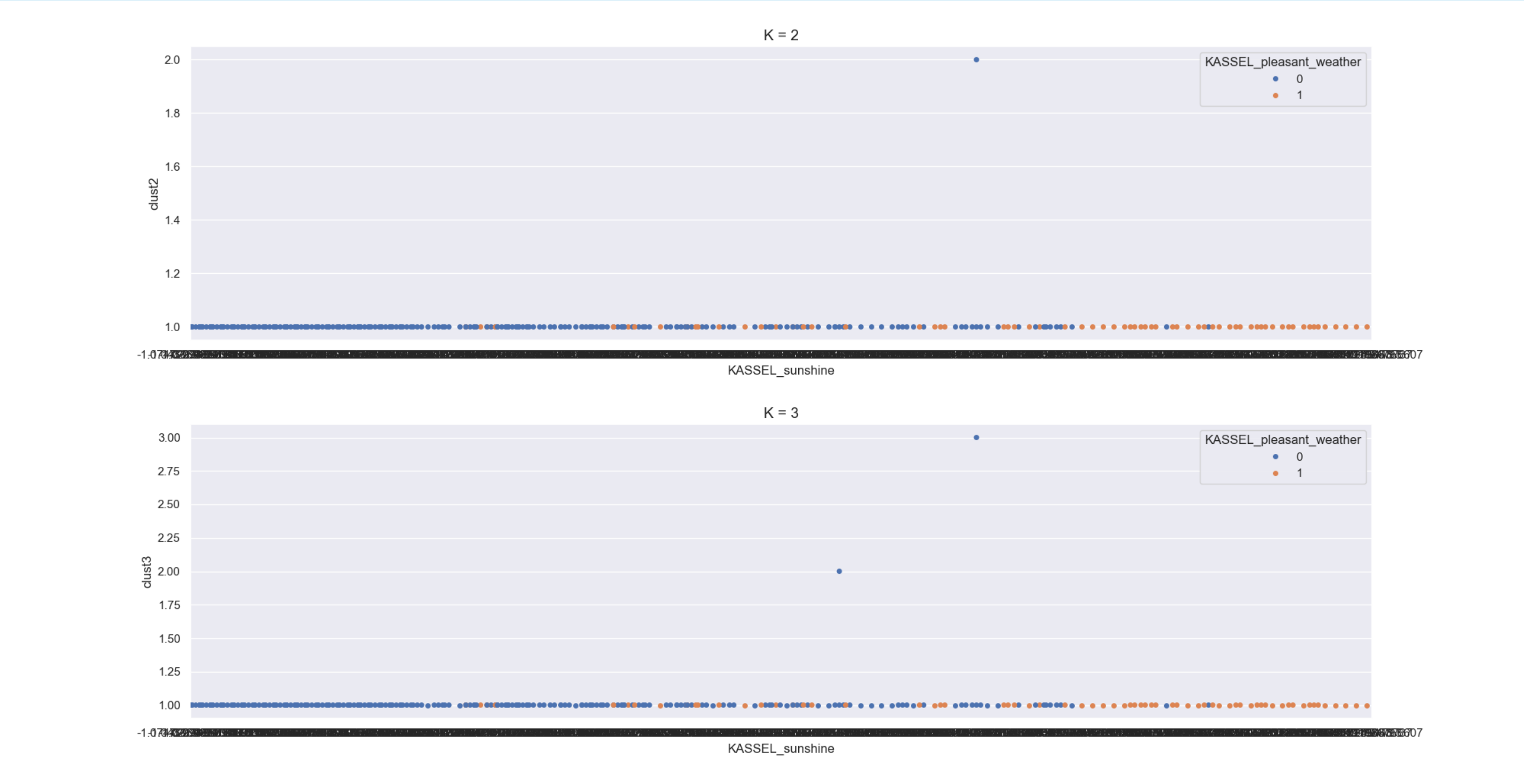
Swarm Plot: Single

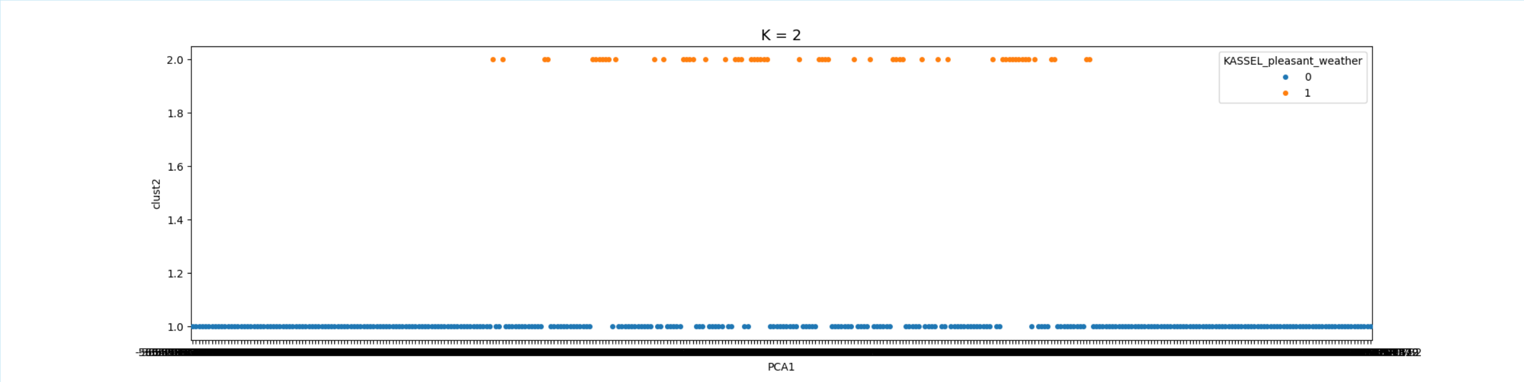
**Top 2 are from original data

Bottom is from PCA**

When examining the swarm plots the single swarm plot plots a single line with rare plot points in their own sections.

The swarm plots that used the PCA data were not insightful.





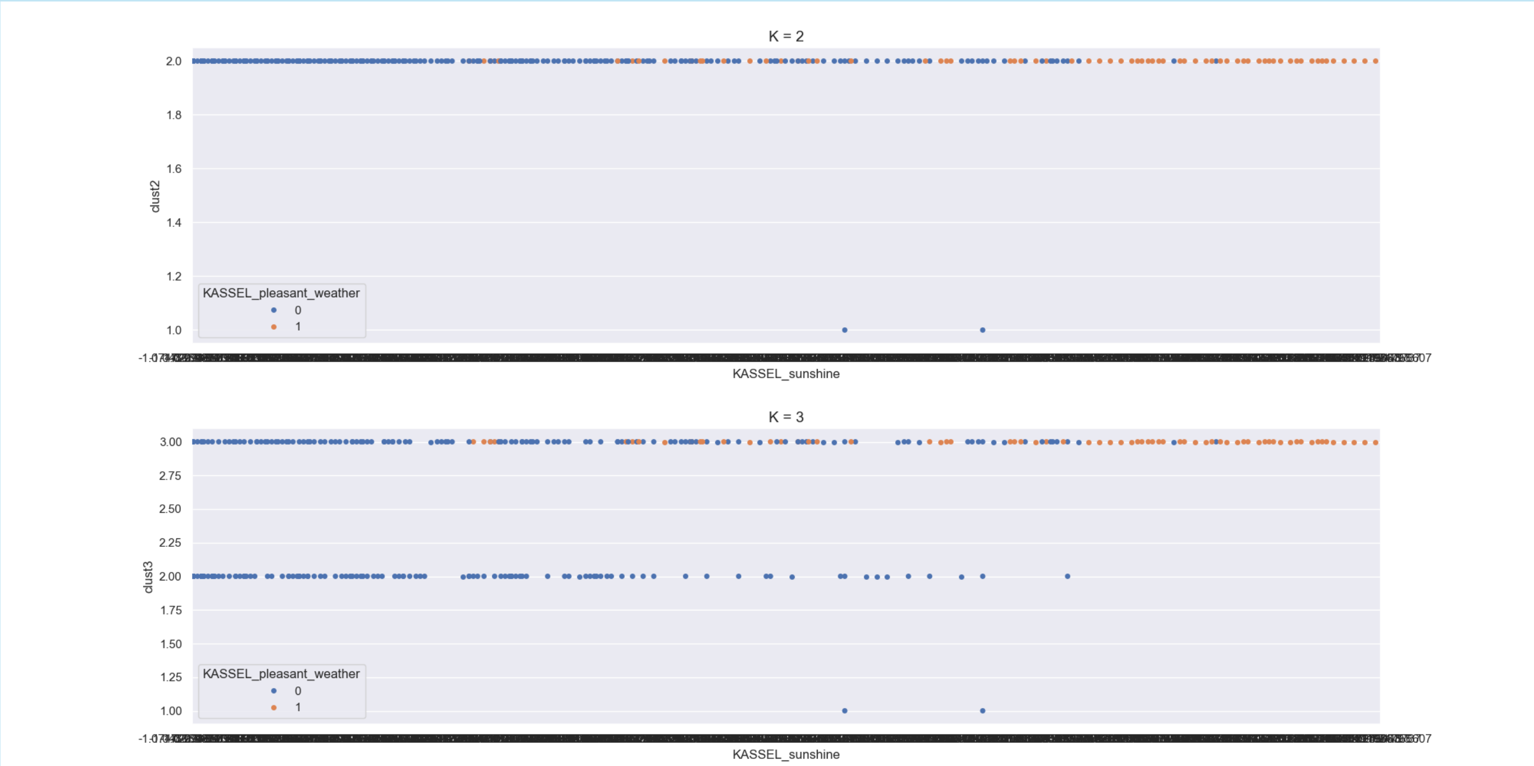
Swarm Plot: Complete

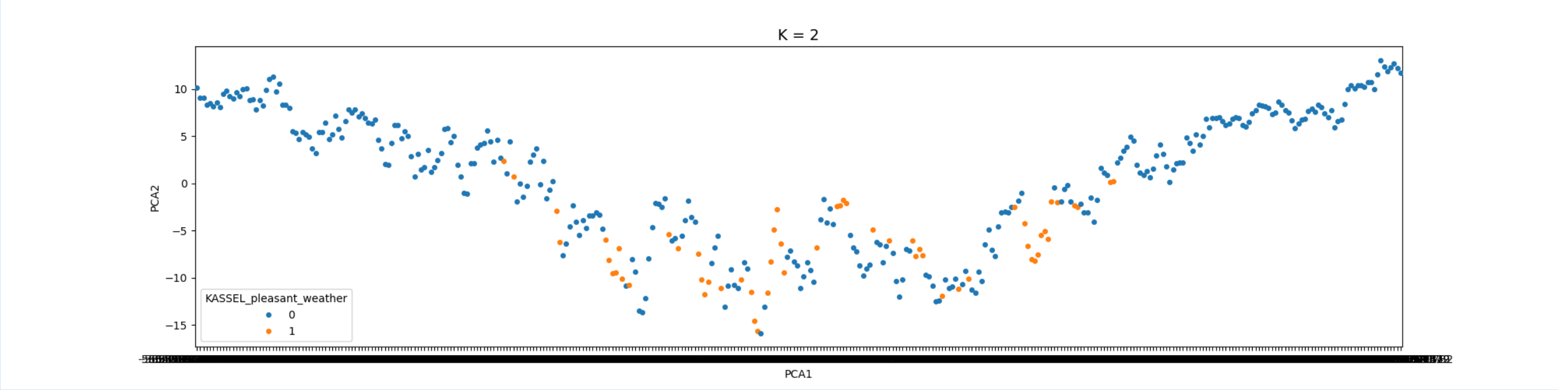
**Top 2 are from original data

Bottom is from PCA**

The complete swarm plot does better work a separating in 3 cluster but it's 2 clusters are poor.

The swarm plots that used the PCA data were not insightful. The completed, average and ward were all identical.

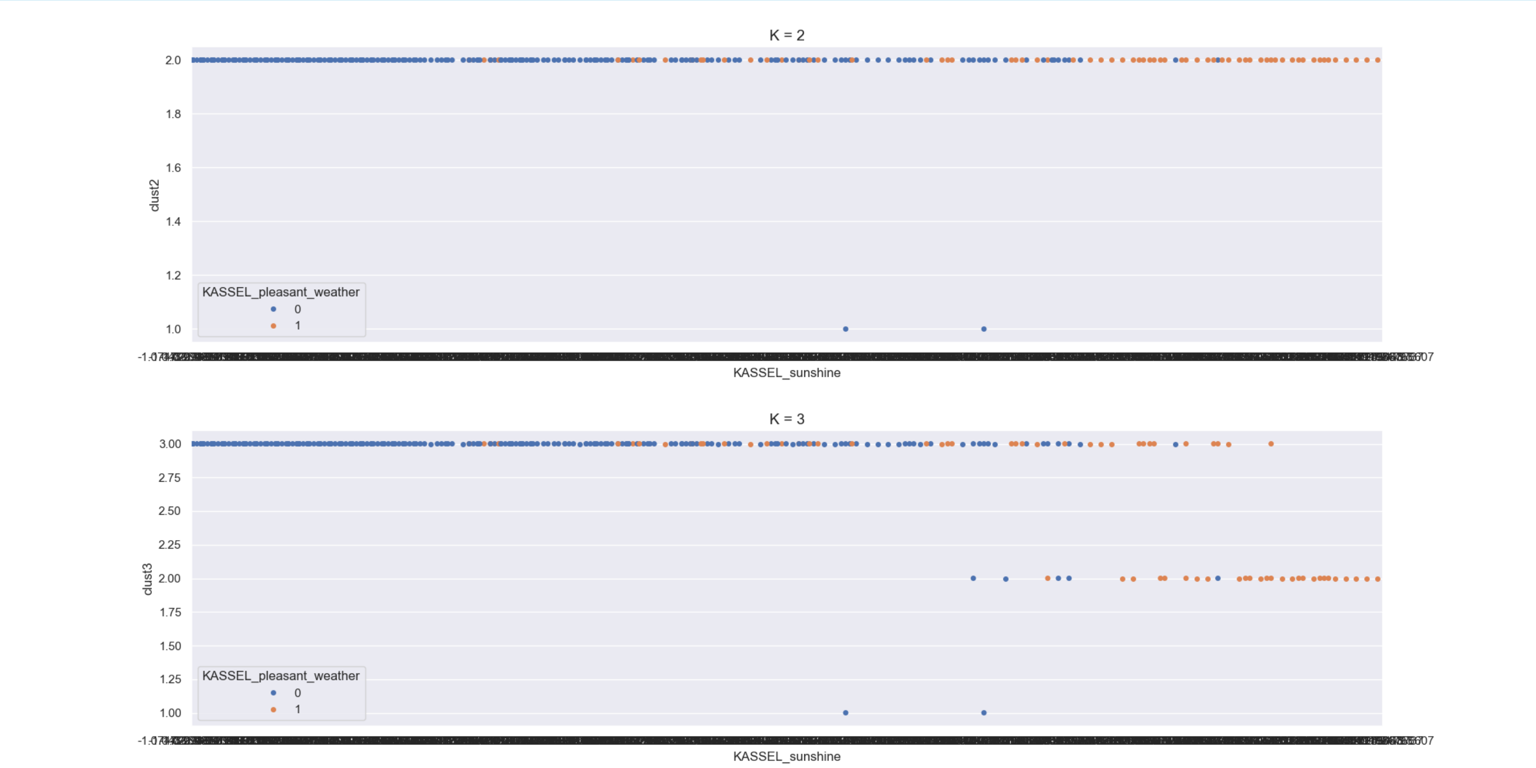


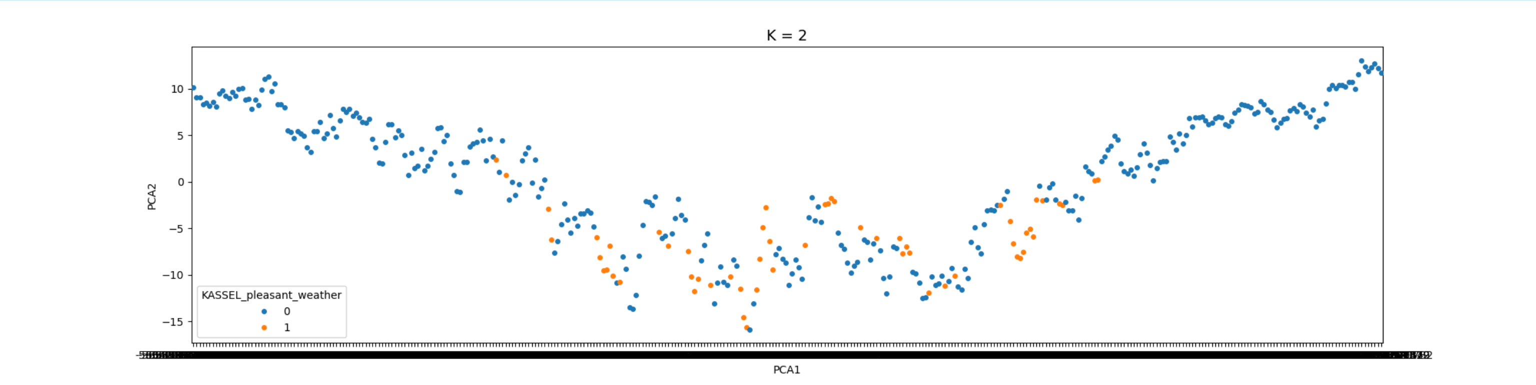


Swarm Plot: Average

**Top 2 are from original data

Bottom is from PCA**





The average swarm plot has similar plots to the complete swarm plot.

The swarm plots that used the PCA data were not insightful. The completed, average and ward were all identical.

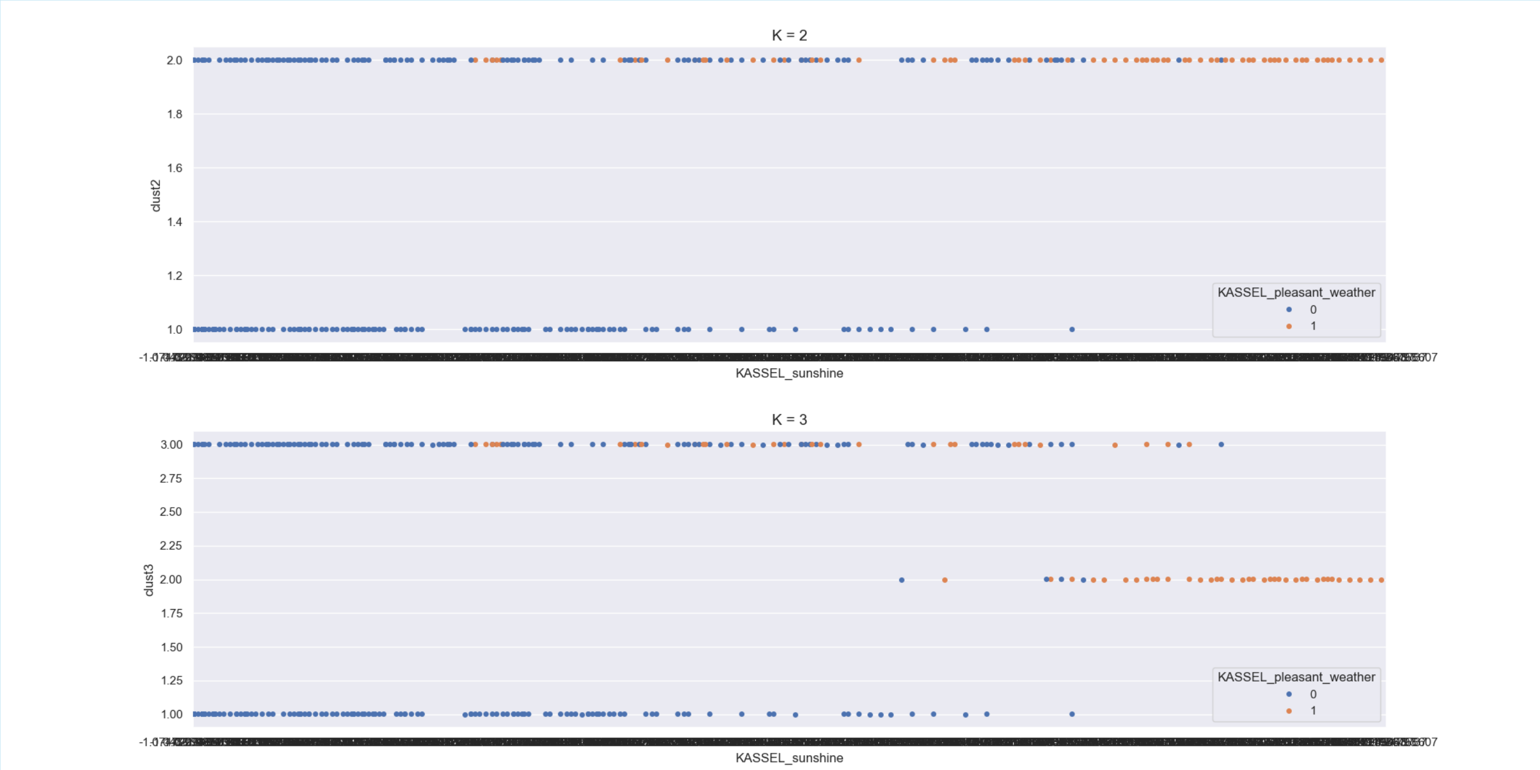
Swarm Plot: Ward

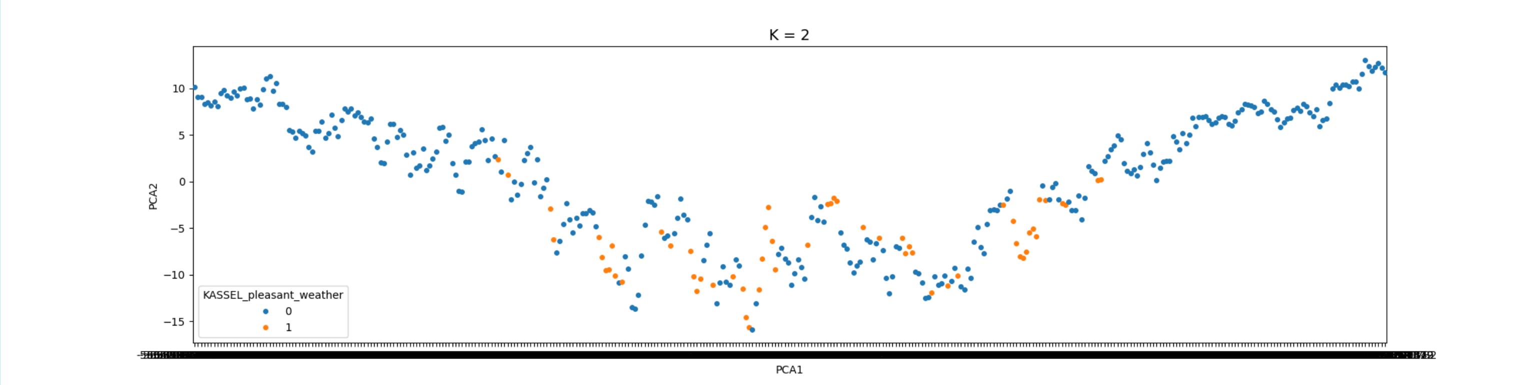
**Top 2 are from original data

Bottom is from PCA**

The ward swarm plot has better separation between the pleasant and unpleasant weather in both its clusters.

The swarm plots that used the PCA data were not insightful. The completed, average and ward were all identical.





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